

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2024 P2H Worked Solutions

Nov 2024 | P2H

31 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3**TOPIC: **Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

1 Show that $1\frac{5}{7} \times 2\frac{3}{16} = 3\frac{3}{4}$

(Total for Question 1 is 3 marks)



PAST PAPER SOLUTION**Q#: 1****Marks: 3**TOPIC: **Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Algebraic proof. The tag is acceptable because the question asks to show the exact result.

Solution

$$1\frac{5}{7} = \frac{12}{7}, \quad 2\frac{3}{16} = \frac{35}{16}$$

$$1\frac{5}{7} \times 2\frac{3}{16} = \frac{12}{7} \times \frac{35}{16}$$

$$= \frac{420}{112} = \frac{15}{4} = 3\frac{3}{4}$$

FINAL ANSWER

$$3\frac{3}{4}.$$

PAST PAPER QUESTION**Q#: 2****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2024 P2H - Nov 2024 | P2H

2 The length of a table is measured as 1.4 metres correct to one decimal place.

(a) Write down the upper bound of the length of the table.

..... metres
(1)

(b) Write down the lower bound of the length of the table.

..... metres
(1)

(Total for Question 2 is 2 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rounding, Estimation and Bounds. The tag is correct.**Method**

The length is 1.4 m correct to one decimal place.

Half of 0.1 is 0.05.

$$1.4 + 0.05 = 1.45$$

$$1.4 - 0.05 = 1.35$$

FINAL ANSWER

(a) 1.45 m, (b) 1.35 m

PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2024 P2H - Nov 2024 | P2H

- 3 The diagram shows triangle PQR

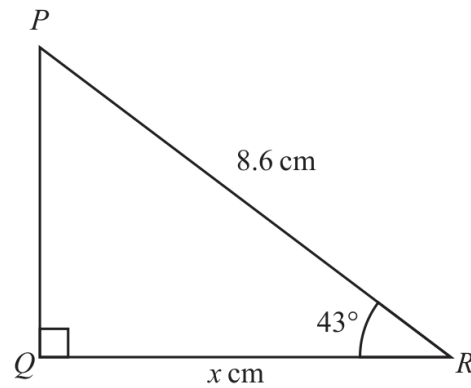


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to Right-Angled Triangles - Pythagoras & Trigonometry.**Solution**

In the right-angled triangle, x is adjacent to the 43° angle.

$$\cos 43^\circ = \frac{x}{8.6}$$

$$x = 8.6 \cos 43^\circ = 6.289 \dots$$

FINAL ANSWER

6.3 cm.

PAST PAPER QUESTION**Q#: 4****Marks: 5**TOPIC: **Percentages**

Nov 2024 P2H - Nov 2024 | P2H

- 4 N is a number.
17% of N is 357
- (a) Work out the value of N

$$N = \dots\dots\dots (2)$$

In 2019, the population of a village was 650
In 2020, the population of the village was 806

- (b) Work out the percentage increase in the population.

$$\dots\dots\dots \% (3)$$

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 5**TOPIC: **Percentages**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Percentages. The tag is correct.**Method**

For part (a),

$$17\% \text{ of } N = 357$$

$$0.17N = 357$$

$$N = \frac{357}{0.17} = 2100$$

For part (b), the increase in population is

$$806 - 650 = 156$$

Percentage increase:

$$\frac{156}{650} \times 100 = 24$$

FINAL ANSWER(a) $N = 2100$, (b) 24%

PAST PAPER QUESTION**Q#: 5****Marks: 4****TOPIC: Probability Toolkit**

Nov 2024 P2H - Nov 2024 | P2H

- 5** Cody has a biased 5-sided spinner, numbered 1, 2, 3, 4, 5

The table gives the probabilities that when the spinner is spun it will land on 2 or on 3 or on 5

Number	1	2	3	4	5
Probability		0.14	0.17		0.21

The probability that the spinner will land on 1 is the same as the probability that the spinner will land on 4

Cody is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on 4

.....
(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 4**TOPIC: **Probability Toolkit**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Probability Toolkit. The tag is correct.**Solution**

The known probabilities add to

$$0.18 + 0.17 + 0.23 = 0.58$$

So the probabilities for 1 and 4 add to

$$1 - 0.58 = 0.42$$

They are equal, so

$$P(4) = 0.42 \div 2 = 0.21$$

Expected number of 4s:

$$400 \times 0.21 = 84$$

FINAL ANSWER

84.

PAST PAPER QUESTIONTOPIC: **Volume & Surface Area**

Nov 2024 P2H - Nov 2024 | P2H

Q#: 6**Marks: 3**

- 6 The diagram shows a solid triangular prism.

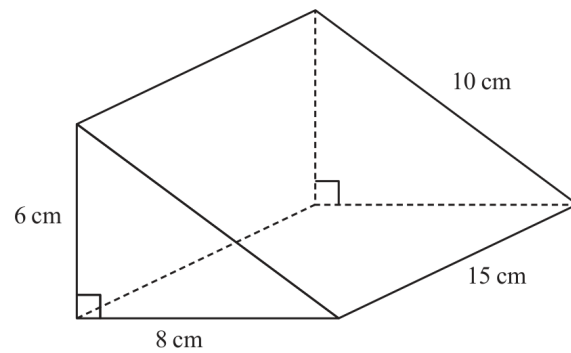


Diagram **NOT**
accurately drawn

Work out the **total** surface area of the triangular prism.

..... cm^2

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 3**TOPIC: **Volume & Surface Area**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Volume & Surface Area. The tag is correct.**Solution**

The triangular end is a right triangle with sides 6, 8, and 10.

Area of the two triangular ends:

$$2 \left(\frac{1}{2} \times 6 \times 8 \right) = 48$$

Area of the three rectangles:

$$15(6 + 8 + 10) = 360$$

Total surface area:

$$48 + 360 = 408$$

FINAL ANSWER408 cm².

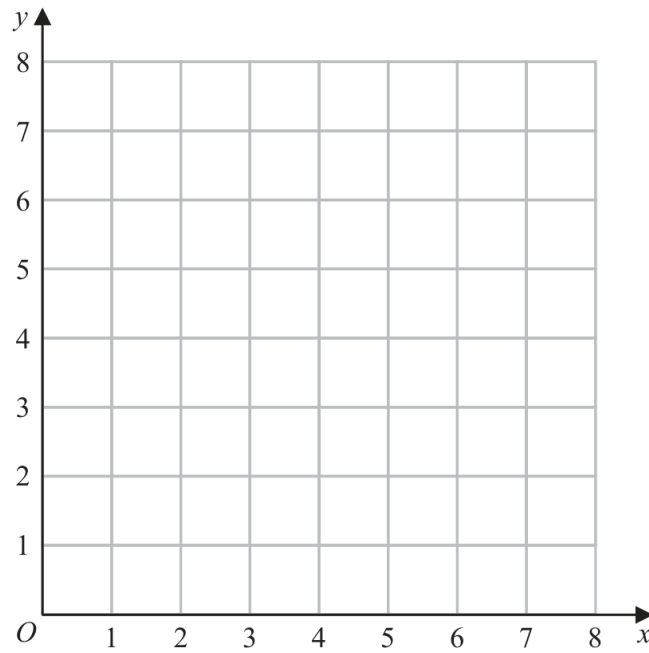
PAST PAPER QUESTION**Q#: 7****Marks: 4****TOPIC: Graphing Inequalities**

Nov 2024 P2H - Nov 2024 | P2H

7 (a) On the grid, draw the straight line with equation

(i) $x = 3$ (ii) $y = 1$ (iii) $x + y = 7$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$x \geq 3 \quad y \geq 1 \quad x + y \leq 7$$

Label the region **R**

(1)

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 4****TOPIC: Graphing Inequalities**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct. This is drawing and shading inequalities.**Solution**

Draw the boundary lines:

$$x = 3, \quad y = 1, \quad x + y = 7$$

The required region is

$$x \geq 3$$

$$y \geq 1$$

$$x + y \leq 7$$

FINAL ANSWERShade $x \geq 3$, $y \geq 1$, $x + y \leq 7$.

PAST PAPER QUESTION**Q#: 8****Marks: 3**TOPIC: **Statistics Toolkit**

Nov 2024 P2H - Nov 2024 | P2H

- 8 Kim puts 4 bananas in a bag.
The mean weight of the 4 bananas in the bag is 145 grams.
- Andy puts one more banana into the bag.
The mean weight of the 5 bananas in the bag is 142 grams.
- Work out the weight of the banana that Andy puts into the bag.

..... grams

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 3**TOPIC: **Statistics Toolkit**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**

The total weight of the first 4 bananas is

$$4 \times 145 = 580$$

The total weight of the 5 bananas is

$$5 \times 142 = 710$$

Weight of the banana Andy added:

$$710 - 580 = 130$$

FINAL ANSWER

130 g.

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Compound Interest & Depreciation**

Nov 2024 P2H - Nov 2024 | P2H

- 9 Nisha invests 20 000 euros for 3 years in a savings account.
She gets 3.5% per year compound interest.

Work out how much money Nisha will have in her savings account
at the end of the 3 years.

Give your answer correct to the nearest euro.

..... euros

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION**Q#: 9****Marks: 3****TOPIC: Compound Interest & Depreciation**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Compound Interest and Depreciation. The tag is correct.**Method**

The multiplier for 3.5% compound interest is

$$1.035$$

After 3 years,

$$20000(1.035)^3 = 22174.3575$$

Correct to the nearest euro,

$$22174.3575 \approx 22174$$

FINAL ANSWER

22174 euros

PAST PAPER QUESTION

Q#: 10

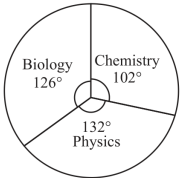
Marks: 5

TOPIC: **Statistics Toolkit**

Nov 2024 P2H - Nov 2024 | P2H

10 All the students in year 10 and all the students in year 11 named their favourite science subject from Biology, Chemistry and Physics.

The pie chart shows information about the results for the year 10 students.
The table shows information about the results for the year 11 students.



Pie chart for year 10

science subject	number of students
Biology	$3x + 6$
Chemistry	$5x + 8$
Physics	$7x - 9$

Table for year 11

There are 300 students in year 10
There are 320 students in year 11

More students in year 10 than in year 11 said Biology was their favourite science subject.

How many more?

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 5**TOPIC: **Statistics Toolkit**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**For year 10, Biology is 126° out of 360° .

$$\frac{126}{360} \times 300 = 105$$

For year 11:

$$(3x + 6) + (5x + 8) + (7x - 9) = 320$$

$$15x + 5 = 320$$

$$x = 21$$

Year 11 Biology:

$$3(21) + 6 = 69$$

Difference:

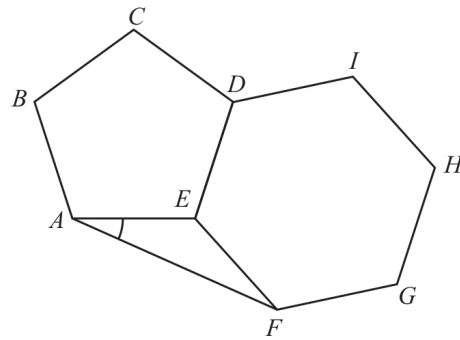
$$105 - 69 = 36$$

FINAL ANSWER

36 more students.

PAST PAPER QUESTION**Q#: 11****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

Nov 2024 P2H - Nov 2024 | P2H

11Diagram **NOT**
accurately drawn

$ABCDE$ is a regular pentagon.
 $DEFGHI$ is a regular hexagon.

AF is a straight line.

Work out the size of angle EAF

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to angles in polygons.**Solution**

A regular pentagon has interior angle

$$108^\circ$$

A regular hexagon has interior angle

$$120^\circ$$

At E , the angle inside triangle AEF is

$$360^\circ - 108^\circ - 120^\circ = 132^\circ$$

Also $AE = ED$ and $ED = EF$, so

$$AE = EF$$

Triangle AEF is isosceles, so the base angles are equal:

$$\angle EAF = \frac{180^\circ - 132^\circ}{2} = 24^\circ$$

FINAL ANSWER 24° .

PAST PAPER QUESTION**Q#: 12****Marks: 7****TOPIC: Rearranging Formulas**

Nov 2024 P2H - Nov 2024 | P2H

- 12 (a) Solve $\frac{3x+2}{5} - \frac{2x+1}{3} = x$
Show clear algebraic working.

$$x = \dots\dots\dots (3)$$

- (b) Make c the subject of the formula $f = \sqrt{\frac{a+bc}{c-d}}$

$$\dots\dots\dots (4)$$

(Total for Question 12 is 7 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 7****TOPIC: Rearranging Formulas**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rearranging formulas. The tag is correct.**Solution**

(a)

$$\frac{3x+2}{5} - \frac{2x+1}{3} = x$$

Multiply by 15:

$$3(3x+2) - 5(2x+1) = 15x$$

$$9x+6 - 10x - 5 = 15x$$

$$1 = 16x$$

$$x = \frac{1}{16}$$

(b)

$$f = \sqrt{\frac{a+bc}{c-d}}$$

$$f^2 = \frac{a+bc}{c-d}$$

$$f^2(c-d) = a+bc$$

$$c(f^2 - b) = a + df^2$$

FINAL ANSWER

$$(a) x = \frac{1}{16}, (b) c = \frac{a+df^2}{f^2-b}$$

PAST PAPER QUESTION

Q#: 13 Marks: 6

TOPIC: Cumulative Frequency Diagrams

Nov 2024 P2H - Nov 2024 | P2H

13 The frequency table gives information about the weights, in kilograms, of 60 parcels in a delivery van.

Weight (<i>w</i> kilograms)	Frequency
$0 < w \leq 1$	4
$1 < w \leq 2$	15
$2 < w \leq 3$	20
$3 < w \leq 4$	11
$4 < w \leq 5$	6
$5 < w \leq 6$	4

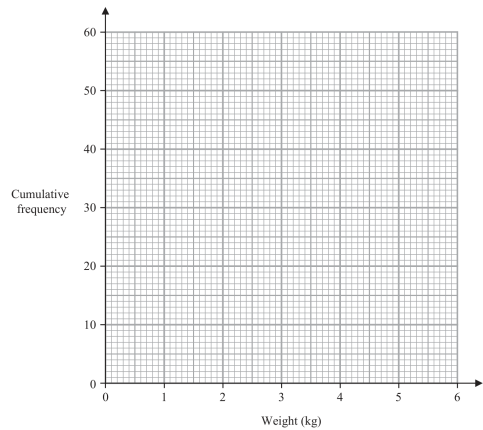
(a) Complete the cumulative frequency table.

Weight (<i>w</i> kilograms)	Cumulative frequency
$0 < w \leq 1$	
$0 < w \leq 2$	
$0 < w \leq 3$	
$0 < w \leq 4$	
$0 < w \leq 5$	
$0 < w \leq 6$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)



(c) Use your graph to find an estimate for the median weight of the 60 parcels.

_____ kilograms
(1)

(d) Use your graph to find an estimate for the number of these parcels that weigh more than 3.7 kilograms.

(2)

(Total for Question 13 is 6 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 6****TOPIC: Cumulative Frequency Diagrams**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Cumulative Frequency Diagrams. The tag is correct.**Solution**

The cumulative frequencies are

$$4, 19, 39, 50, 56, 60$$

The median is the 30th value.

$$2 + \frac{30 - 19}{39 - 19} \times 1 = 2.55$$

At 3.7 kg,

$$F(3.7) \approx 39 + \frac{0.7}{1}(50 - 39) = 46.7$$

So the number heavier than 3.7 kg is

$$60 - 46.7 = 13.3$$

FINAL ANSWER

median about 2.55 kg, and about 13 parcels.

PAST PAPER QUESTION**Q#: 14****Marks: 3**TOPIC: **Algebraic Roots & Indices**

Nov 2024 P2H - Nov 2024 | P2H

14 Given that $\frac{3^{2n+3}}{3^4} = 3^3 \times 3^{1-2n}$

find the value of n

Show your working clearly.

$$n = \dots\dots\dots$$

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 3**TOPIC: **Algebraic Roots & Indices**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Algebraic roots and indices. The tag is correct.**Solution**

$$\frac{3^{2n+3}}{3^4} = 3^{2n-1}$$

$$3^3 \times 3^{1-2n} = 3^{4-2n}$$

So

$$2n - 1 = 4 - 2n$$

$$4n = 5$$

FINAL ANSWER

$$n = \frac{5}{4}$$

PAST PAPER QUESTION**Q#: 15****Marks: 2**TOPIC: **Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

15 Use algebra to show that $0.7\dot{6}\dot{3} = \frac{42}{55}$

(Total for Question 15 is 2 marks)

Dr. Eslam

PAST PAPER SOLUTION**Q#: 15****Marks: 2**TOPIC: **Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Algebraic proof. The tag is correct.**Solution**

Let

$$x = 0.7636363 \dots$$

Then

$$10x = 7.636363 \dots$$

and

$$1000x = 763.636363 \dots$$

Subtract:

$$990x = 756$$

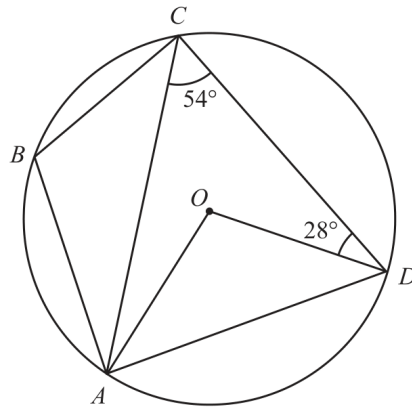
$$x = \frac{756}{990} = \frac{42}{55}$$

FINAL ANSWER

$$0.7636363 \dots = \frac{42}{55}.$$

PAST PAPER QUESTION**Q#: 16****Marks: 5**TOPIC: **Circle Theorems**

Nov 2024 P2H - Nov 2024 | P2H

16 A, B, C and D are points on a circle, centre O Diagram **NOT**
accurately drawn(a) (i) Work out the size of angle AOD

.....^o
(1)

(ii) Give a reason for your answer to part (a)(i)

.....
.....
(1)

(b) Work out the size of angle CAO

.....^o
(1)

(c) Work out the size of angle ABC

.....^o
(2)

(Total for Question 16 is 5 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 5****TOPIC: Circle Theorems**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Circle Theorems. The tag is correct.**Solution**(a) The angle at the centre is twice the angle at the circumference standing on the same arc AD :

$$\angle AOD = 2 \times 54^\circ = 108^\circ$$

(b) Since $OC = OD$, triangle COD is isosceles:

$$\angle OCD = \angle CDO = 28^\circ$$

So

$$\angle COD = 180^\circ - 28^\circ - 28^\circ = 124^\circ$$

Angles around O add to 360° :

$$\angle AOC = 360^\circ - 108^\circ - 124^\circ = 128^\circ$$

Since $OA = OC$,

$$\angle CAO = \frac{180^\circ - 128^\circ}{2} = 26^\circ$$

(c) Angle ABC stands on the major arc AC , so

$$\angle ABC = \frac{360^\circ - 128^\circ}{2} = 116^\circ$$

FINAL ANSWER $\angle AOD = 108^\circ$, $\angle CAO = 26^\circ$, and $\angle ABC = 116^\circ$.

PAST PAPER QUESTION**Q#: 17****Marks: 3**TOPIC: **Vectors**

Nov 2024 P2H - Nov 2024 | P2H

17 Here are two vectors.

$$\vec{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix} \quad \vec{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$$

Calculate the magnitude of the vector $\vec{HF}$

.....

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 3**TOPIC: **Vectors**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Vectors. The tag is correct.**Solution**

$$\overrightarrow{GF} = -\overrightarrow{FG} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$$

$$\overrightarrow{HF} = \overrightarrow{HG} + \overrightarrow{GF} = \begin{pmatrix} 4 \\ 14 \end{pmatrix} + \begin{pmatrix} 5 \\ -2 \end{pmatrix} = \begin{pmatrix} 9 \\ 12 \end{pmatrix}$$

Magnitude:

$$|\overrightarrow{HF}| = \sqrt{9^2 + 12^2} = 15$$

FINAL ANSWER

15.

PAST PAPER QUESTION**Q#: 18****Marks: 4**TOPIC: **Linear Graphs ($y = mx + c$)**

Nov 2024 P2H - Nov 2024 | P2H

- 18** The straight line **L** is perpendicular to the straight line with equation $2x + y = 9$ and passes through the point with coordinates (8, 11)

Find an equation for **L**

Give your answer in the form $y = mx + c$

.....
(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 4****TOPIC: Linear Graphs ($y = mx + c$)**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to linear graphs. The question asks for the equation of a perpendicular line.

Solution

The given line is

$$2x + y = 9$$

$$y = -2x + 9$$

Its gradient is -2 , so a perpendicular line has gradient $\frac{1}{2}$.

Line L passes through $(8, 11)$:

$$y - 11 = \frac{1}{2}(x - 8)$$

$$y = \frac{1}{2}x + 7$$

FINAL ANSWER

$$y = \frac{1}{2}x + 7.$$

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Transformations of Graphs**

Nov 2024 P2H - Nov 2024 | P2H

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are (5, 4)

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 5)$

(.....,)
(1)

(ii) $y = 3f(x)$

(.....,)
(1)

(iii) $y = f(x) - 7$

(.....,)
(1)**(Total for Question 19 is 3 marks)**

PAST PAPER SOLUTION**Q#: 19****Marks: 3****TOPIC: Transformations of Graphs**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to transformations of graphs. The question asks for transformed minimum points of $y = f(x)$.

Solution

The minimum point of $y = f(x)$ is

$$(5, 4)$$

(i) For

$$y = f(x + 5)$$

the graph moves 5 units left, so the minimum point becomes

$$(0, 4)$$

(ii) For

$$y = 3f(x)$$

the y-coordinate is multiplied by 3, so the minimum point becomes

$$(5, 12)$$

(iii) For

$$y = f(x) - 7$$

the graph moves 7 units down, so the minimum point becomes

$$(5, -3)$$

FINAL ANSWER

$(0, 4)$, $(5, 12)$, and $(5, -3)$.

PAST PAPER QUESTION**Q#: 20****Marks: 3**TOPIC: **Solving Inequalities**

Nov 2024 P2H - Nov 2024 | P2H

- 20** Solve the inequality $10x^2 + 11x - 21 < 0$
Show clear algebraic working.

(Total for Question 20 is 3 marks)

Dr. Eslam

PAST PAPER SOLUTION**Q#: 20****Marks: 3**TOPIC: **Solving Inequalities**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$10x^2 + 11x - 21 < 0$$

Factorise:

$$10x^2 + 11x - 21 = (10x + 21)(x - 1)$$

The critical values are

$$x = -\frac{21}{10} \quad \text{and} \quad x = 1$$

The quadratic is less than 0 between the roots.

FINAL ANSWER

$$-\frac{21}{10} < x < 1$$

PAST PAPER QUESTION**Q#: 20****Marks: 3**TOPIC: **Solving Inequalities**

Nov 2024 P2H - Nov 2024 | P2H

- 20** Solve the inequality $10x^2 + 11x - 21 < 0$
Show clear algebraic working.

(Total for Question 20 is 3 marks)

Dr. Eslam

PAST PAPER SOLUTION**Q#: 20****Marks: 3**TOPIC: **Solving Inequalities**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$10x^2 + 11x - 21 < 0$$

Factorise:

$$10x^2 + 11x - 21 = (10x + 21)(x - 1)$$

The critical values are

$$x = -\frac{21}{10} \quad \text{and} \quad x = 1$$

The quadratic is less than 0 between the roots.

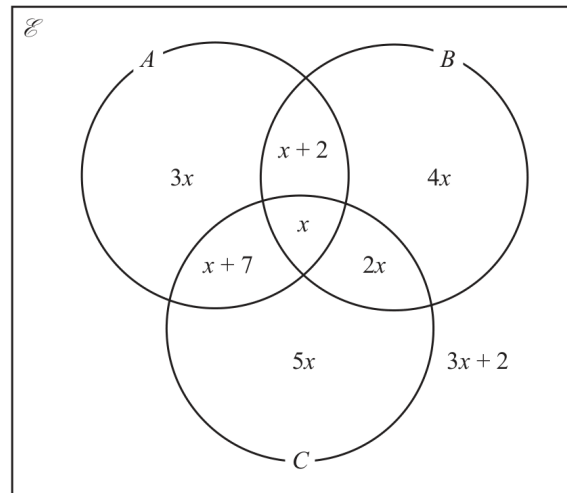
FINAL ANSWER

$$-\frac{21}{10} < x < 1$$

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Set Notation & Venn Diagrams**

Nov 2024 P2H - Nov 2024 | P2H

- 21 The Venn diagram shows information about the numbers of items in set A , set B and set C , where x is an integer.



Given that $n(A \cup B)' = 26$

find $n(A' \cap C)$

$$n(A' \cap C) = \dots\dots\dots$$

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 4****TOPIC: Set Notation & Venn Diagrams**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Set notation and Venn diagrams. The tag is correct.

Method

The complement of $A \cup B$ contains the regions outside both A and B . These are $5x$ and $3x + 2$.
So $5x + (3x + 2) = 26$, hence $8x + 2 = 26$ and $x = 3$.

Now $A' \cap C$ means inside C , but not inside A . These are the regions $5x$ and $2x$.

So $n(A' \cap C) = 5x + 2x = 7x = 7(3) = 21$.

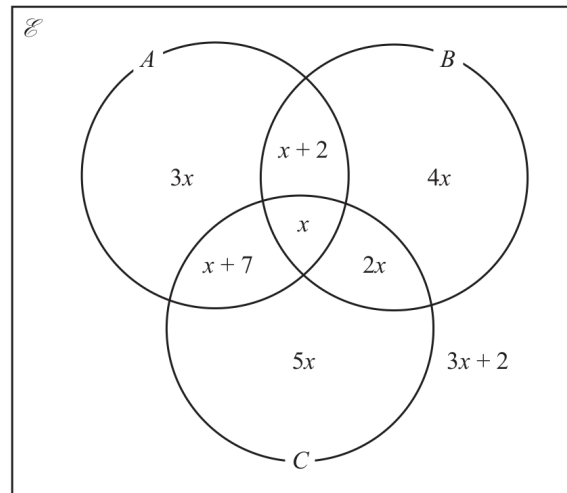
FINAL ANSWER

21

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Set Notation & Venn Diagrams**

Nov 2024 P2H - Nov 2024 | P2H

- 21** The Venn diagram shows information about the numbers of items in set A , set B and set C , where x is an integer.



Given that $n(A \cup B)' = 26$

find $n(A' \cap C)$

$$n(A' \cap C) = \dots\dots\dots$$

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 4****TOPIC: Set Notation & Venn Diagrams**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Set notation and Venn diagrams. The tag is correct.**Method**

The complement of $A \cup B$ contains the regions outside both A and B . These are $5x$ and $3x + 2$.
So $5x + (3x + 2) = 26$, hence $8x + 2 = 26$ and $x = 3$.

Now $A' \cap C$ means inside C , but not inside A . These are the regions $5x$ and $2x$.

So $n(A' \cap C) = 5x + 2x = 7x = 7(3) = 21$.

FINAL ANSWER

21

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

Nov 2024 P2H - Nov 2024 | P2H

22 Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$x^2 + y^2 + y = 3$$

$$y = x + 2$$

Substitute:

$$x^2 + (x + 2)^2 + (x + 2) = 3$$

$$x^2 + x^2 + 4x + 4 + x + 2 = 3$$

$$2x^2 + 5x + 3 = 0$$

$$(2x + 3)(x + 1) = 0$$

So

$$x = -\frac{3}{2} \quad \text{or} \quad x = -1$$

Find y :

$$x = -\frac{3}{2} \Rightarrow y = \frac{1}{2}$$

$$x = -1 \Rightarrow y = 1$$

FINAL ANSWER

$$(x, y) = \left(-\frac{3}{2}, \frac{1}{2}\right) \text{ or } (-1, 1)$$

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

Nov 2024 P2H - Nov 2024 | P2H

22 Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$x^2 + y^2 + y = 3$$

$$y = x + 2$$

Substitute:

$$x^2 + (x + 2)^2 + (x + 2) = 3$$

$$x^2 + x^2 + 4x + 4 + x + 2 = 3$$

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So

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Find y :

$$x = -\frac{3}{2} \Rightarrow y = \frac{1}{2}$$

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FINAL ANSWER

$$(x, y) = \left(-\frac{3}{2}, \frac{1}{2}\right) \text{ or } (-1, 1)$$

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

23 Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 4**TOPIC: **Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x)$$

Factorise:

$$16x^2 - 36 = 4(2x - 3)(2x + 3)$$

$$2x^2 + 7x + 6 = (2x + 3)(x + 2)$$

$$x^2 - 5x - 14 = (x - 7)(x + 2)$$

So

$$\begin{aligned} & \frac{4(2x - 3)(2x + 3)}{x - 7} \div \frac{(2x + 3)(x + 2)}{(x - 7)(x + 2)} - (7 + 8x) \\ &= \frac{4(2x - 3)(2x + 3)}{x - 7} \times \frac{(x - 7)(x + 2)}{(2x + 3)(x + 2)} - (7 + 8x) \\ &= 4(2x - 3) - (7 + 8x) \\ &= 8x - 12 - 7 - 8x \\ &= -19 \end{aligned}$$

So $n = -19$, which is an integer.**FINAL ANSWER**

$$n = -19$$

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

23 Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 4**TOPIC: **Algebraic Proof**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x)$$

Factorise:

$$16x^2 - 36 = 4(2x - 3)(2x + 3)$$

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So

$$\begin{aligned} & \frac{4(2x - 3)(2x + 3)}{x - 7} \div \frac{(2x + 3)(x + 2)}{(x - 7)(x + 2)} - (7 + 8x) \\ &= \frac{4(2x - 3)(2x + 3)}{x - 7} \times \frac{(x - 7)(x + 2)}{(2x + 3)(x + 2)} - (7 + 8x) \\ &= 4(2x - 3) - (7 + 8x) \\ &= 8x - 12 - 7 - 8x \\ &= -19 \end{aligned}$$

So $n = -19$, which is an integer.**FINAL ANSWER**

$$n = -19$$

PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Bearings, Scale Drawing & Constructions**

Nov 2024 P2H - Nov 2024 | P2H

24 A , B and C are three points on horizontal ground. $AB = 8.4$ metres $BC = 9.2$ metres B is on a bearing of 067° from A C is on a bearing of 129° from B Calculate the bearing of A from C

Give your answer correct to the nearest degree.

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 6****TOPIC: Bearings, Scale Drawing & Constructions**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

Resolve each journey into east and north components.

From A to B :

$$\text{east} = 8.4 \sin 67^\circ = 7.7322 \dots$$

$$\text{north} = 8.4 \cos 67^\circ = 3.2821 \dots$$

From B to C :

$$\text{east} = 9.2 \sin 129^\circ = 7.1497 \dots$$

$$\text{north} = 9.2 \cos 129^\circ = -5.7897 \dots$$

So from A to C :

$$\text{east} = 14.8820 \dots$$

$$\text{north} = -2.5076 \dots$$

Therefore from C to A :

$$\text{east} = -14.8820 \dots, \quad \text{north} = 2.5076 \dots$$

This direction is west of north.

$$\theta = \tan^{-1} \left(\frac{14.8820 \dots}{2.5076 \dots} \right) = 80.4355 \dots^\circ$$

Bearing of A from C :

$$360^\circ - 80.4355 \dots^\circ = 279.5644 \dots^\circ$$

FINAL ANSWER 280°

PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Bearings, Scale Drawing & Constructions**

Nov 2024 P2H - Nov 2024 | P2H

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PAST PAPER SOLUTION**Q#: 24****Marks: 6****TOPIC: Bearings, Scale Drawing & Constructions**

Nov 2024 P2H - Nov 2024 | P2H

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From B to C :

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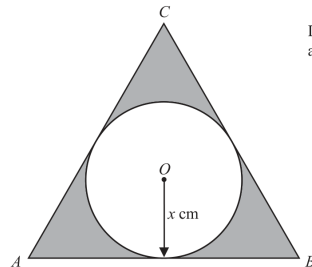
Bearing of A from C :

$$360^\circ - 80.4355 \dots^\circ = 279.5644 \dots^\circ$$

FINAL ANSWER 280°

PAST PAPER QUESTION**Q#: 25****Marks: 5****TOPIC: Circles, Arcs & Sectors**

Nov 2024 P2H - Nov 2024 | P2H

25 The diagram shows an equilateral triangle ABC and a circle with centre O Diagram **NOT**
accurately drawn AB , BC and CA are tangents to the circle.The radius of the circle is x cmThe total area, in cm^2 , of the regions shown shaded in the diagram is nx^2 Find the value of n

Give your answer correct to 3 significant figures.

 $n = \dots\dots\dots$

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION**Q#: 25****Marks: 5****TOPIC: Circles, Arcs & Sectors**

Nov 2024 P2H - Nov 2024 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved to Circles, Arcs & Sectors.**Method**

The circle has radius x , so the inradius of the equilateral triangle is x .
 For an equilateral triangle with side length a ,

$$\text{inradius} = \frac{a\sqrt{3}}{6}$$

So

$$x = \frac{a\sqrt{3}}{6}$$

$$a = \frac{6x}{\sqrt{3}} = 2\sqrt{3}x$$

Area of the triangle:

$$\begin{aligned} \frac{\sqrt{3}}{4}a^2 &= \frac{\sqrt{3}}{4}(2\sqrt{3}x)^2 \\ &= \frac{\sqrt{3}}{4}(12x^2) = 3\sqrt{3}x^2 \end{aligned}$$

Area of the circle:

$$\pi x^2$$

The shaded area is

$$\begin{aligned} &3\sqrt{3}x^2 - \pi x^2 \\ &= (3\sqrt{3} - \pi)x^2 \end{aligned}$$

Therefore

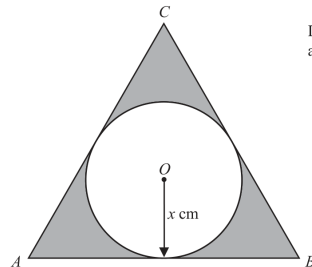
$$n = 3\sqrt{3} - \pi$$

FINAL ANSWER

$$3\sqrt{3} - \pi$$

PAST PAPER QUESTION**Q#: 25****Marks: 5****TOPIC: Circles, Arcs & Sectors**

Nov 2024 P2H - Nov 2024 | P2H

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Give your answer correct to 3 significant figures.

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PAST PAPER SOLUTION**Q#: 25****Marks: 5****TOPIC: Circles, Arcs & Sectors**

Nov 2024 P2H - Nov 2024 | P2H

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The shaded area is

$$\begin{aligned} &3\sqrt{3}x^2 - \pi x^2 \\ &= (3\sqrt{3} - \pi)x^2 \end{aligned}$$

Therefore

$$n = 3\sqrt{3} - \pi$$

FINAL ANSWER

$$3\sqrt{3} - \pi$$